

GERSON RAMOS

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PROFESSIONAL SUMMARY

Data Scientist and ML Developer with over 5 years of experience bridging advanced analytics, software engineering, and financial strategy. Expert in Python and Advanced SQL, with a proven track record in building and deploying Classic ML (XGBoost, Random Forest), Deep Learning, and Generative AI solutions. Currently architecting large-scale cloud migrations (GCP to AWS) for enterprise environments (Stellantis). Proficient in the AWS ecosystem (SageMaker, Bedrock, Glue) and dedicated to creating scalable, high-accuracy models that drive business intelligence.

PROFESSIONAL EXPERIENCE

Data Scientist / Cloud Data Engineer – CompassUOL (Client: Stellantis) *Nov 2024 – Present / Remote*

Migration Tracking Dashboard

Built a comprehensive interactive dashboard for Stellantis system migration tracking using Python, Streamlit and Plotly, replacing manual spreadsheet-based processes and centralizing strategic and operational KPIs in real time for managers and technical teams.

Architecture: Modular architecture with separation between data layer (ETL via Jira API), calculation layer, and visualization layer with pages and tabs per user profile, following maintainability and testability best practices.

Authentication: Authentication system with profile-based access control (executive, strategic, admin) integrated with Supabase.

Data Pipeline: Automated Jira data extraction pipeline with scheduled updates via GitHub Actions four times per day.

Indicators: Weighted progress indicators by delivery cycle stage (Development, Homologation, Pre-Production, Assisted Production) with planned calculation based on real epic deadlines.

Defect Management: Tracking of Storybugs, Technical Debts and Bugs with cycle time in business days, SLA alerts, and distribution by Data-Lake and owner (Compass vs. Stellantis).

Visualizations: Burn-up charts, CFD (Cumulative Flow Diagram), Lead Time, weekly Throughput, and probabilistic delivery forecasting via Monte Carlo simulation.

Executive View: Executive dashboard with consolidated progress view, epic Gantt chart, total delivery delta vs. planned, and homologation and assisted production charts by Data-Lake.

Interface: Light and dark theme support with responsiveness for use in management presentations.

Stack: Python 3.14, Streamlit, Plotly, Pandas, Jira REST API, Supabase, GitHub Actions, Git.

GCP to AWS Data Pipeline Migration (multi-domain)

Led the migration of data pipelines across multiple business domains from GCP to AWS, ensuring full result parity between environments and operational continuity throughout the transition.

Migration: Migrated ingestion and transformation pipelines from BigQuery and Cloud Functions to Snowflake and Apache Airflow, faithfully reproducing the original business logic.

Orchestration: Implemented DAGs with EKS Pod Operator orchestration covering multiple data domains with incremental and full-load strategies.

Quality: Identified and resolved circular dependency issues between pipelines, restructuring the orchestration flow to ensure consistency and correct execution order.

Financial Precision: Ensured numeric precision in financial calculation migrations, reproducing BigQuery NUMERIC rounding behavior in Snowflake.

Standardization: Standardized security and infrastructure configurations across dozens of DAGs distributed across multiple domains, reducing inconsistencies and simplifying maintenance.

Modeling: Modeled tables and views across Snowflake data layers (SL, TL, BL) with incremental load strategies by period.

Stack: Snowflake, Apache Airflow, AWS EKS, S3, Python, SQL, BigQuery, Cloud Functions, GitLab CI/CD.

Data Scientist – Outlier AI

Nov 2024 – Present | Remote

Focused on Deep Learning and Generative AI through experimentation and training of Large Language Models (LLMs) for automated reasoning.

GenAI and Deep Learning: Built pipelines for RAG (Retrieval-Augmented Generation) testing and fine-tuning of LLMs. Created synthetic datasets and benchmark evaluation scripts using Python.

Model Optimization: Contributed to automated testing frameworks to evaluate model reasoning capabilities, reporting bugs and improving model accuracy.

Data Engineer – CompassUOL (Project: Pottencial Seguros)

Jun 2024 – Present | Rio de Janeiro, RJ

Implemented a high-performance Medallion Architecture (Bronze, Silver, Gold) to support advanced analytics and ML models.

Performance Engineering: Refactored Airflow orchestrations with custom sensors, reducing worker blockage by 99% and increasing pipeline throughput by 35% using Apache Iceberg and Spark.

Data Pipeline: Built real-time CDC ingestion via Kafka and batch processing with Spark on EMR, ensuring clean data availability for downstream ML applications.

ML and AI Developer – CompassUOL (AI Studio)

Jun 2024 – Nov 2024 | Mendes, RJ

Classic ML: Developed high-accuracy predictive models using XGBoost and Random Forest for hotel pricing and sports outcomes (87% accuracy).

GenAI Integration: Developed serverless APIs integrating Amazon Bedrock and Amazon Lex for Natural Language Processing (NLP) solutions.

Data Scientist / Sr. Data Analyst – Pet Premium Distribuidora

Sep 2020 – Feb 2024 | Rio de Janeiro, RJ

Leveraged statistical modeling and Python to transform financial data into predictive insights.

Predictive Modeling: Developed and deployed sales forecasting models using Python (Scikit-Learn) and linear regression techniques, directly influencing inventory planning.

Advanced Analytics: Conducted deep exploratory data analysis (EDA) using SQL and Python (Pandas/NumPy) to identify market trends.

Business Intelligence: Built dynamic dashboards in Power BI to visualize model outputs and track KPIs for executive decision-making.

TECHNICAL SKILLS

Machine Learning: Classic ML (Regression, Classification, Clustering, XGBoost, Random Forest), Deep Learning (Neural Networks, TensorFlow), Model Evaluation.

Generative AI: LLMs (Large Language Models), RAG, Prompt Engineering, Amazon Bedrock, LangChain.

Languages: Python (Advanced: Pandas, NumPy, Scikit-Learn, TensorFlow), SQL (Advanced), R.

Cloud and Engineering: AWS (SageMaker, Lambda, Glue, EMR, Redshift), GCP (Dataflow, Cloud Functions), Snowflake, Apache Airflow, Spark, Kafka, Docker, Git, GitLab CI/CD.

Visualization and BI: Streamlit, Plotly, Power BI, Jira REST API, Supabase, GitHub Actions.

EDUCATION AND CERTIFICATIONS

Degree: Bachelor's Degree in Data Science, Estacio de Sa (Expected December 2025)

Degree: Bachelor's in Business Administration (Incomplete), Universidade Veiga de Almeida (2015-2018)

Certifications

AWS Certified Cloud Practitioner

GenAI Technical Certification

Google Advanced Data Analytics Certificate

IBM Machine Learning Specialist: Associate

Data Analyst Associate: IBM